

Sub : Cyber Shield

Defend, Investigate, Resolve

From Evidence to Action – Complete Forensic
Solutions.

Presented By :

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Introduction to CYBER SHIELD :

Problem Statement :

Core Issue : Organizations face significant challenges in managing and investigating cybersecurity incidents due to fragmented tools, lack of centralized evidence management, and inefficient case tracking systems. Cyber Shield aims to address the critical gap between incident occurrence and effective forensic investigation by providing a unified platform for cybersecurity professionals to manage cases, evidence, and intelligence efficiently.

Key Problems:

- Disconnected forensic tools requiring multiple platforms
- Inefficient evidence collection and management processes
- Delayed incident response due to poor case organization
- Lack of integrated knowledge base for threat intelligence
- Limited reporting capabilities for compliance and audit purposes



Why Does This Problem Exist?

Root Causes:

1. Technological Fragmentation :

- Multiple vendors providing point solutions
- Incompatible data formats between tools
- Lack of standardized forensic procedures

2. Resource Constraints :

- Shortage of skilled cybersecurity professionals
- Limited budget for comprehensive forensic solutions
- Time pressure during active investigations

3. Organizational Factors :

- Siloed departments handling different security aspects
- Inconsistent incident response protocols
- Poor documentation practices

4. Evolving Threat Landscape :

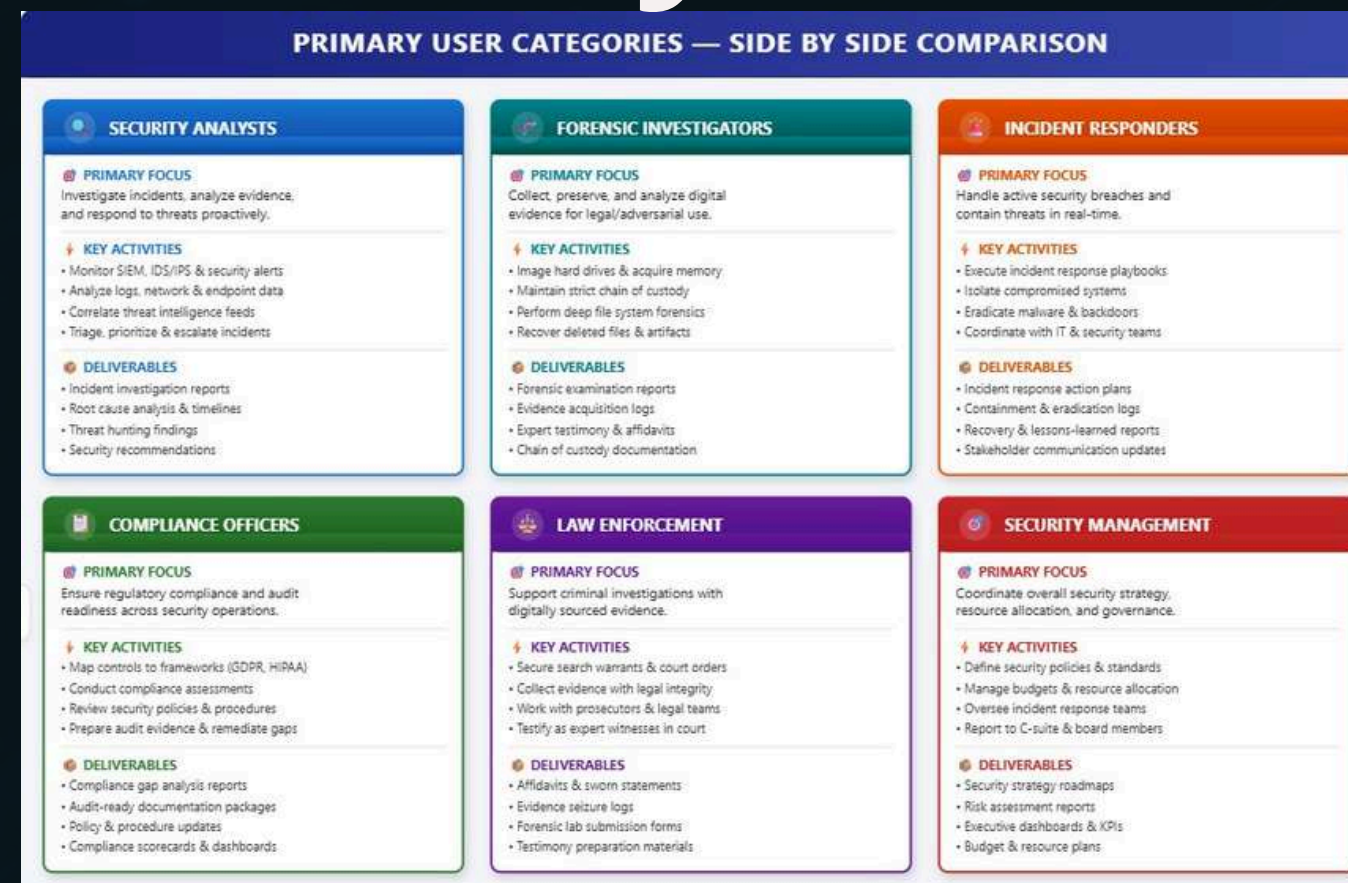
- Rapidly changing attack vectors
- Increasing sophistication of cybercriminals
- Growing volume of digital evidence

5. Regulatory Requirements :

- Complex compliance mandates (GDPR, HIPAA, PCI-DSS)
- Reporting obligations across jurisdictions
- Chain of custody requirements



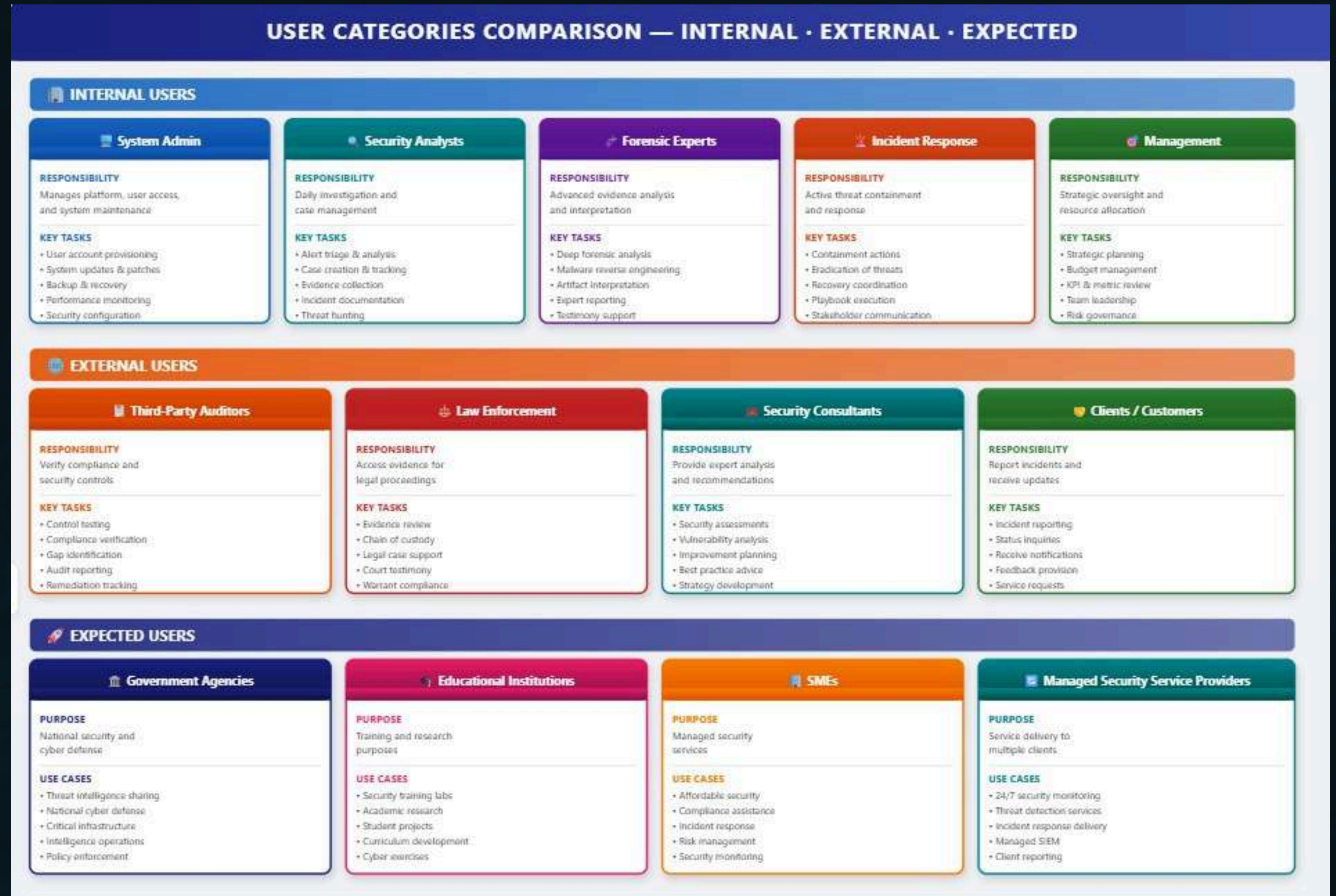
Who Uses Cyber Shield?



Beneficiaries :

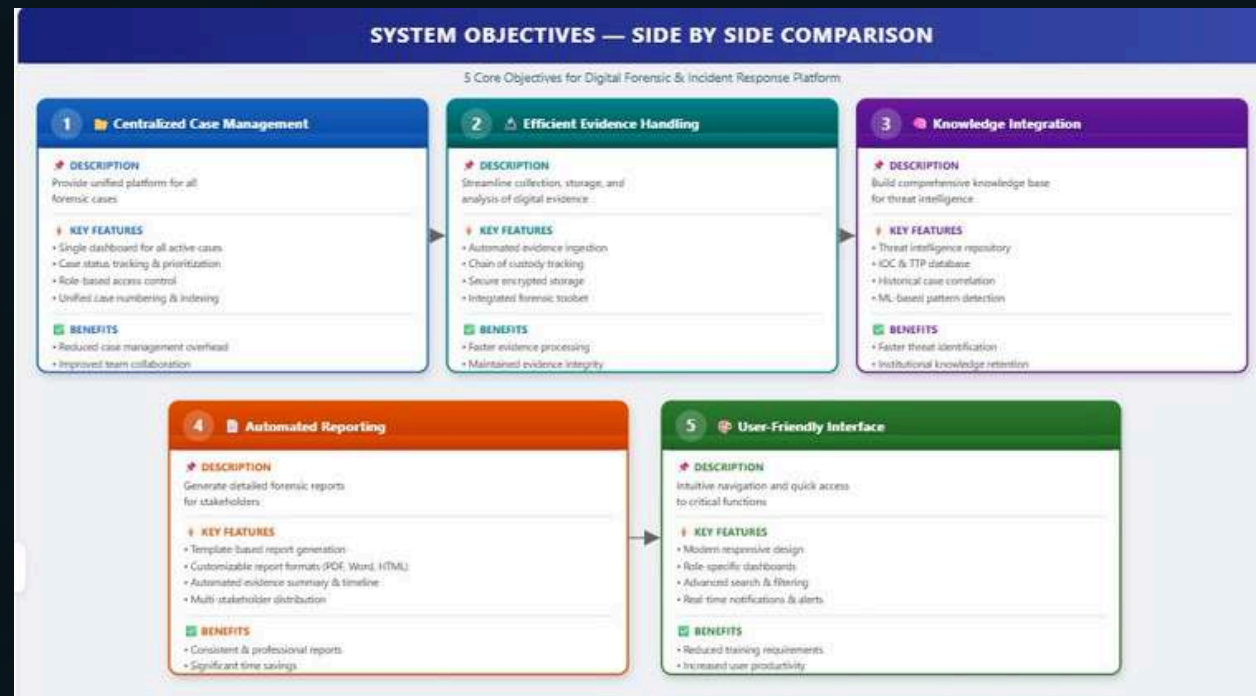
- C-level Executives – Risk management and strategic decisions
- IT Teams – System protection and vulnerability management
- Legal Teams – Evidence handling and litigation support
- Employees – Protection of personal and organizational data
- Customers – Assurance of data security and privacy

Stakeholders :



Objectives of the Project :

Primary Objectives :



Secondary Objectives :

- Reduce incident response time by 40%
- Improve evidence integrity through chain of custody tracking
- Enhance collaboration among team members
- Provide real-time threat alerts and updates
- Ensure regulatory compliance

Expected Outputs :



Hands On Activity – Information Levels :



User and Activity User Interactions And Activities :

Admin

DESCRIPTION

System administrator

KEY ACTIVITIES

- User provisioning
- Access control
- System maintenance
- Backup management
- Configuration mgmt

PERMISSIONS

- Full system access

Forensic Analyst

DESCRIPTION

Investigation lead

KEY ACTIVITIES

- Case creation/mgmt
- Evidence analysis
- Report generation
- Knowledge base update
- Deep forensics

PERMISSIONS

- Full case & evidence

Incident Responder

DESCRIPTION

First responder

KEY ACTIVITIES

- Alert triage
- Initial investigation
- Evidence collection
- Containment actions
- Incident docs

PERMISSIONS

- Case creation & update

Security Auditor

DESCRIPTION

Compliance checker

KEY ACTIVITIES

- Review cases
- Check compliance
- Verify procedures
- Generate audit reports
- Review audit trails

PERMISSIONS

- Read only + audit

Viewer

DESCRIPTION

Read only access

KEY ACTIVITIES

- View dashboards
- Access reports
- Monitor cases
- View metrics
- Review findings

PERMISSIONS

- Read only access

Admin

PRIMARY ACTIVITIES

- User provisioning — Create, modify, and deactivate user accounts
- Access control — Assign roles and manage permissions
- System maintenance — Apply updates and configuration changes
- Backup management — Schedule and verify system backups

SYSTEM INTERACTIONS

- Admin dashboard — System health, user metrics, and alerts
- User management panel — Add, edit, and delete users
- System settings — Configure system parameters
- Audit logs — View system activity and security events

Forensic Analyst

PRIMARY ACTIVITIES

- Case creation/management — Create, assign, and track cases
- Evidence upload/analysis — Upload and analyze digital evidence
- Report generation — Create detailed forensic reports
- Knowledge base update — Add findings and best practices

SYSTEM INTERACTIONS

- Case module — View, create, and manage forensic cases
- Evidence manager — Upload, store, and analyze evidence
- Report generator — Create and export professional reports
- Knowledge base — Search and update threat intelligence

Incident Responder

PRIMARY ACTIVITIES

- Alert triage — Prioritize and assign security alerts
- Initial investigation — Gather initial threat information
- Evidence collection — Collect and preserve volatile data
- Containment actions — Isolate and contain active threats

SYSTEM INTERACTIONS

- Alert dashboard — View and manage security alerts
- Quick case creation — Create incident cases from alerts
- Evidence upload — Upload collected evidence to system
- Case updates — Document actions and findings

Security Auditor

PRIMARY ACTIVITIES

- Review cases — Examine case documentation and evidence
- Check compliance — Verify adherence to policies and regulations
- Verify procedures — Ensure proper investigation methodology
- Generate audit reports — Create compliance and audit reports

SYSTEM INTERACTIONS

- Case viewer — Read-only access to case details
- Report viewer — Access and review forensic reports
- Audit trail — View system and case audit logs
- Compliance reports — Generate audit and compliance reports

Viewer

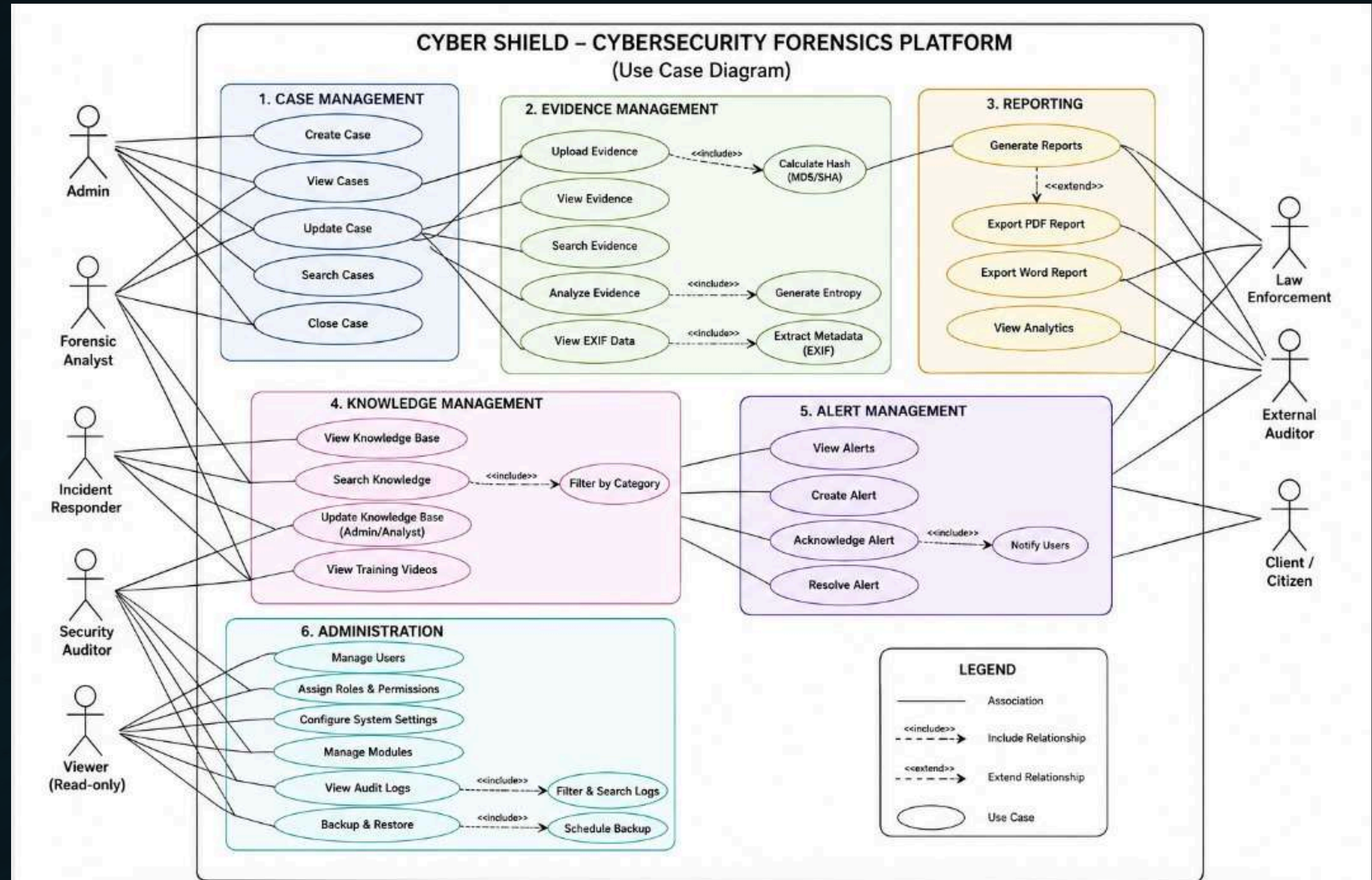
PRIMARY ACTIVITIES

- View dashboards — Monitor case and system metrics
- Access reports — Read and download forensic reports
- Monitor cases — Track case progress and status
- View metrics — Review performance and trend data

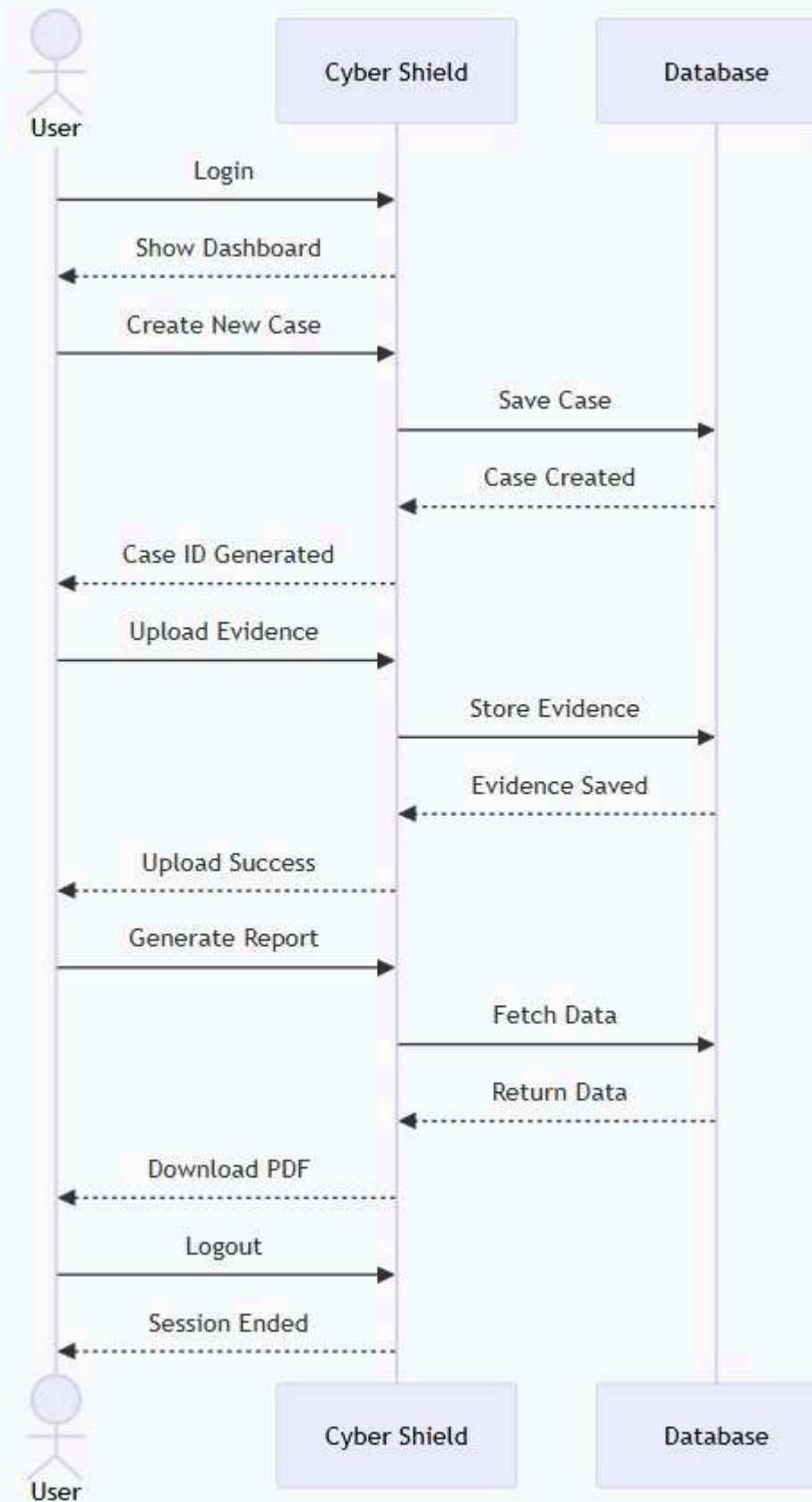
SYSTEM INTERACTIONS

- Read-only dashboard — View limited dashboard metrics
- Report viewer — Access reports in read-only mode
- Case monitor — Track case status without editing
- Metrics viewer — View pre-configured metrics

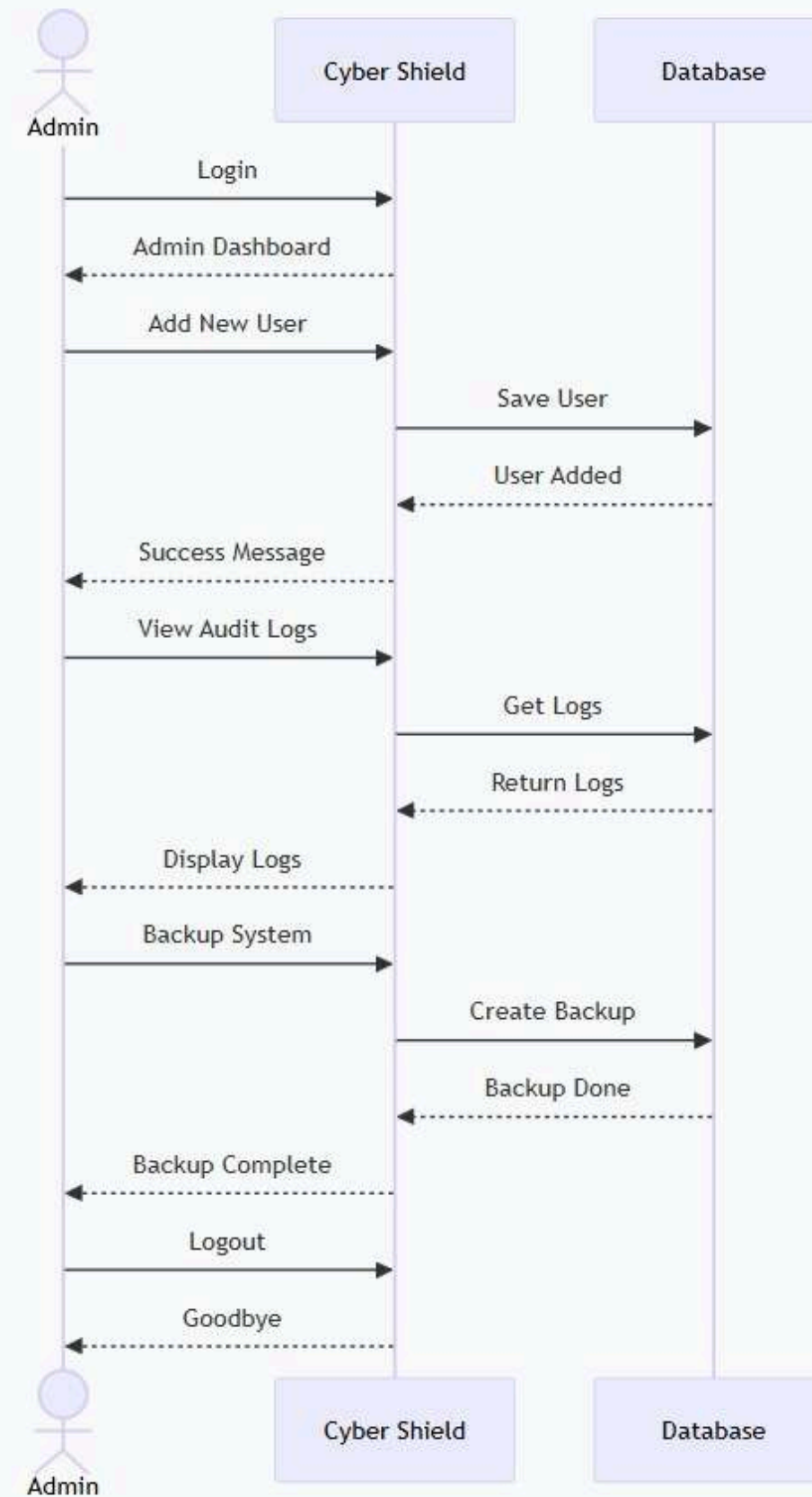
Use Case Diagram :



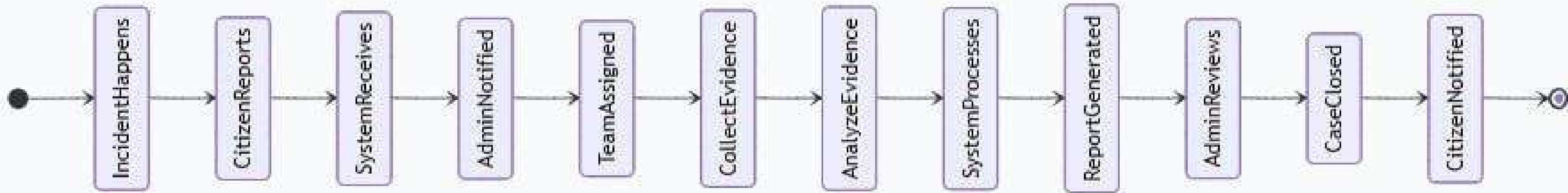
User Activity Diagram :



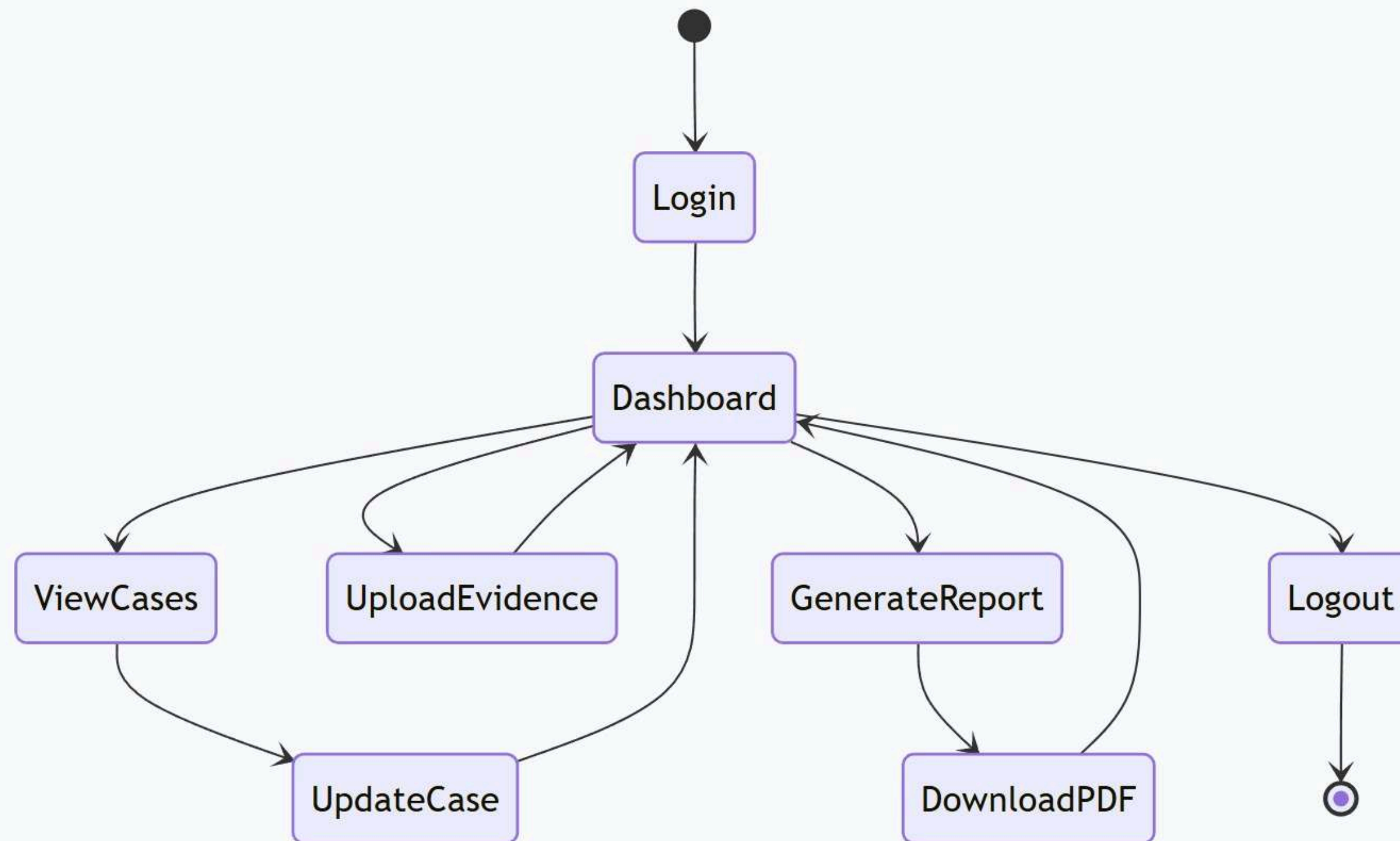
Admin Activity Diagram :



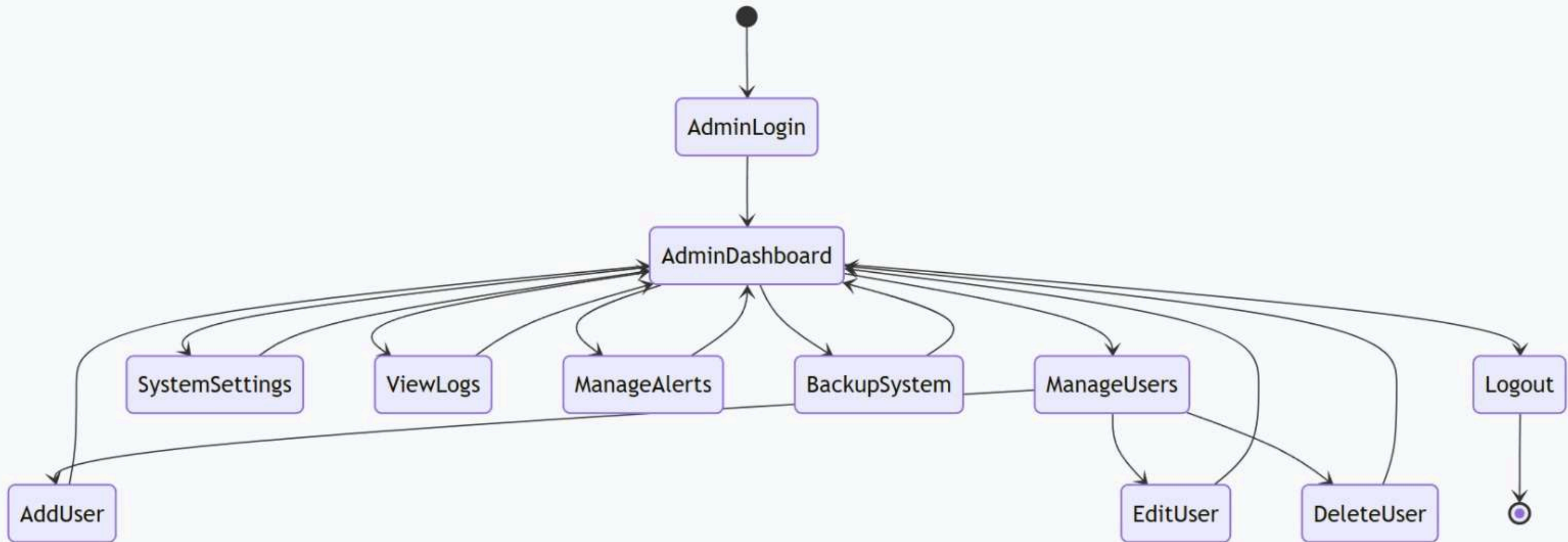
Activity Diagram – Citizen, System, Admin, Rescue Team :



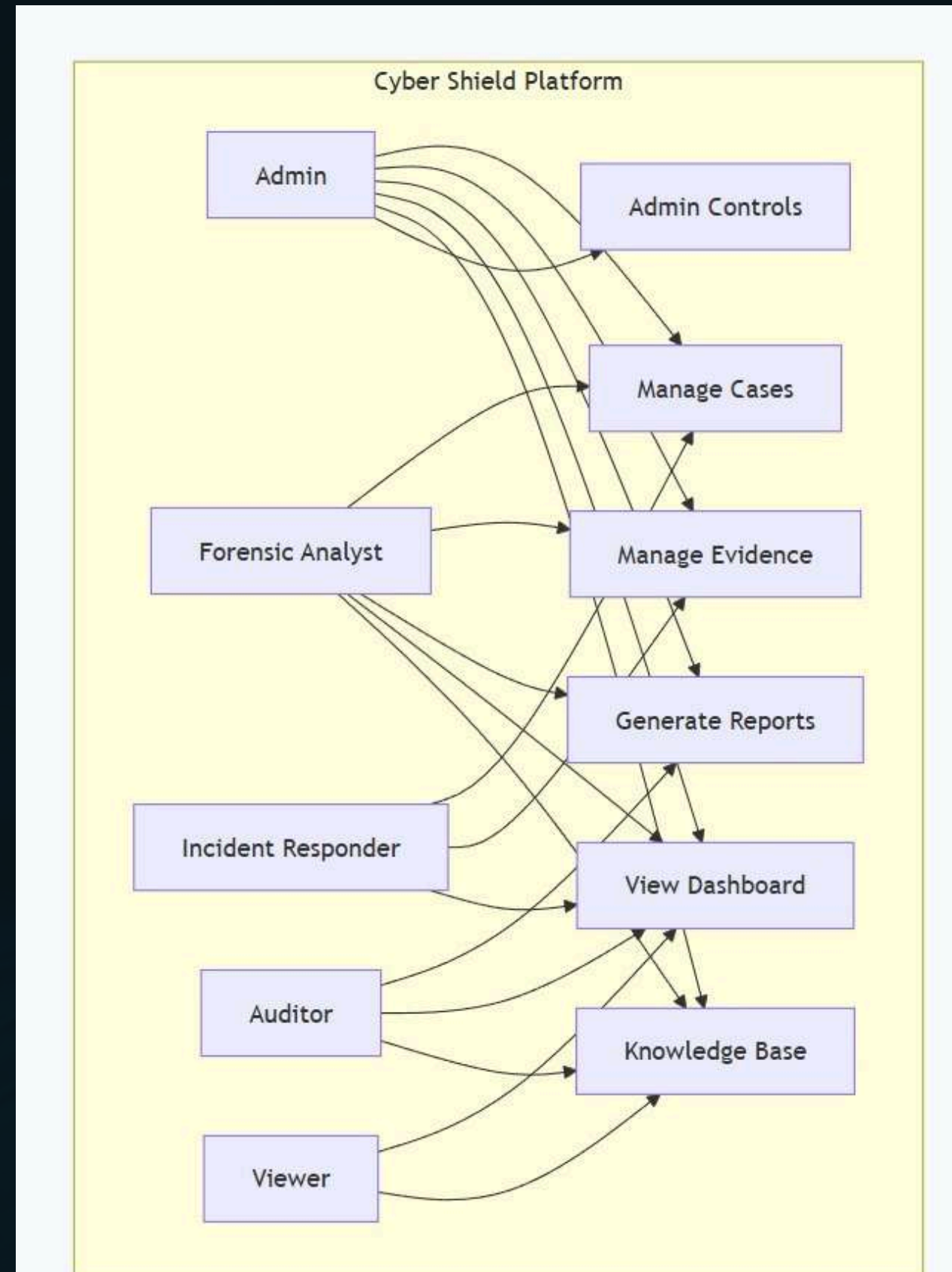
User Sequence Diagram - Cyber Shield :



Admin Sequence Diagram – Cyber Shield :



UML Communication Diagram :



Summary Of Findings :

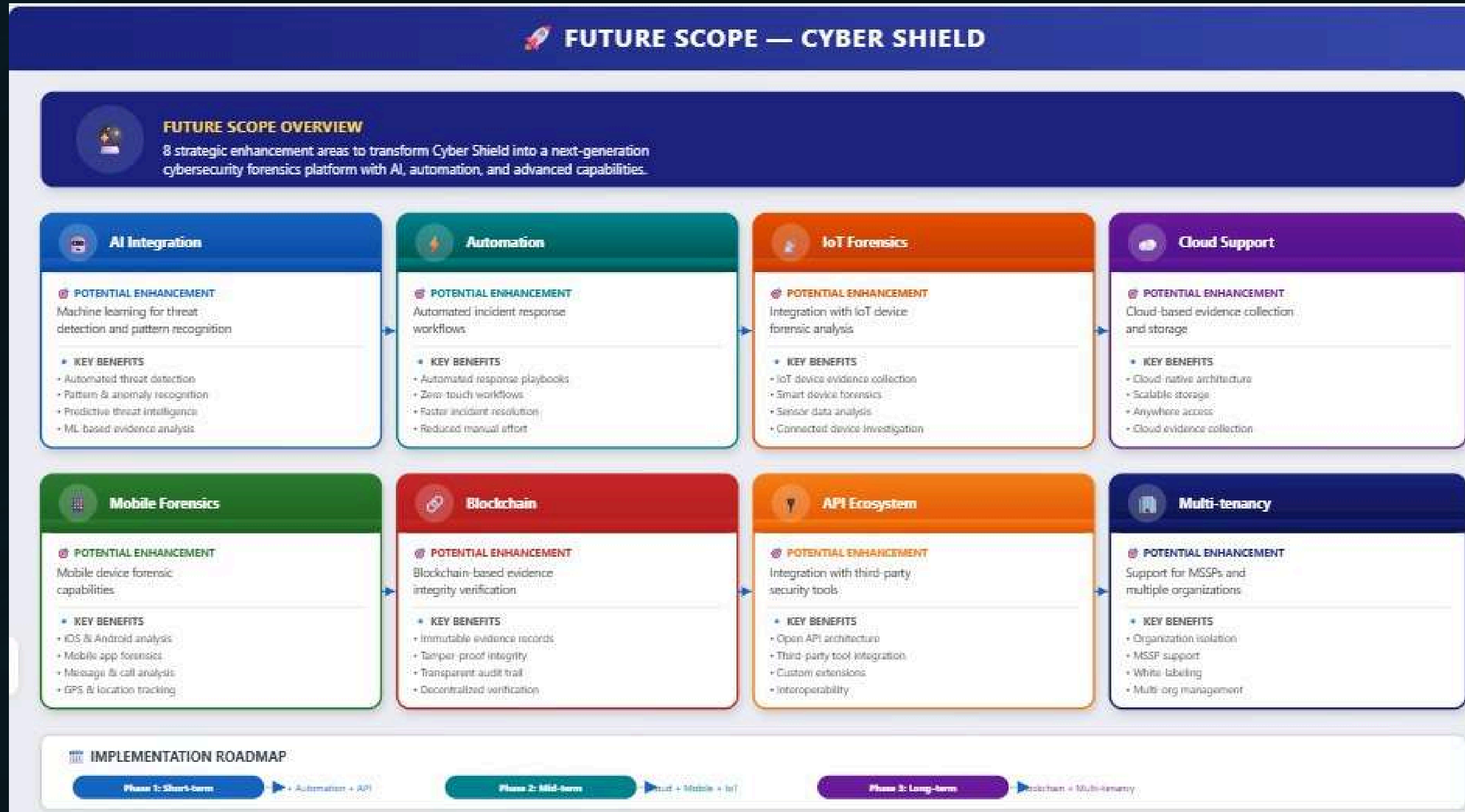


Key Outcomes :

- Improved Incident Response
- Reduction in case resolution time
- Better evidence preservation and chain of custody
- Enhanced team collaboration
- Enhanced Decision Making
- Real-time analytics and dashboards
- Strategic information for management
- Tactical insights for forensic teams
- Comprehensive Evidence Management
- Secure storage with integrity verification
- Advanced analysis capabilities
- Easy retrieval and search
- Knowledge Integration
- Centralized threat intelligence
- Training and awareness resources
- Best practices documentation



Future Scope :



Final Remarks :

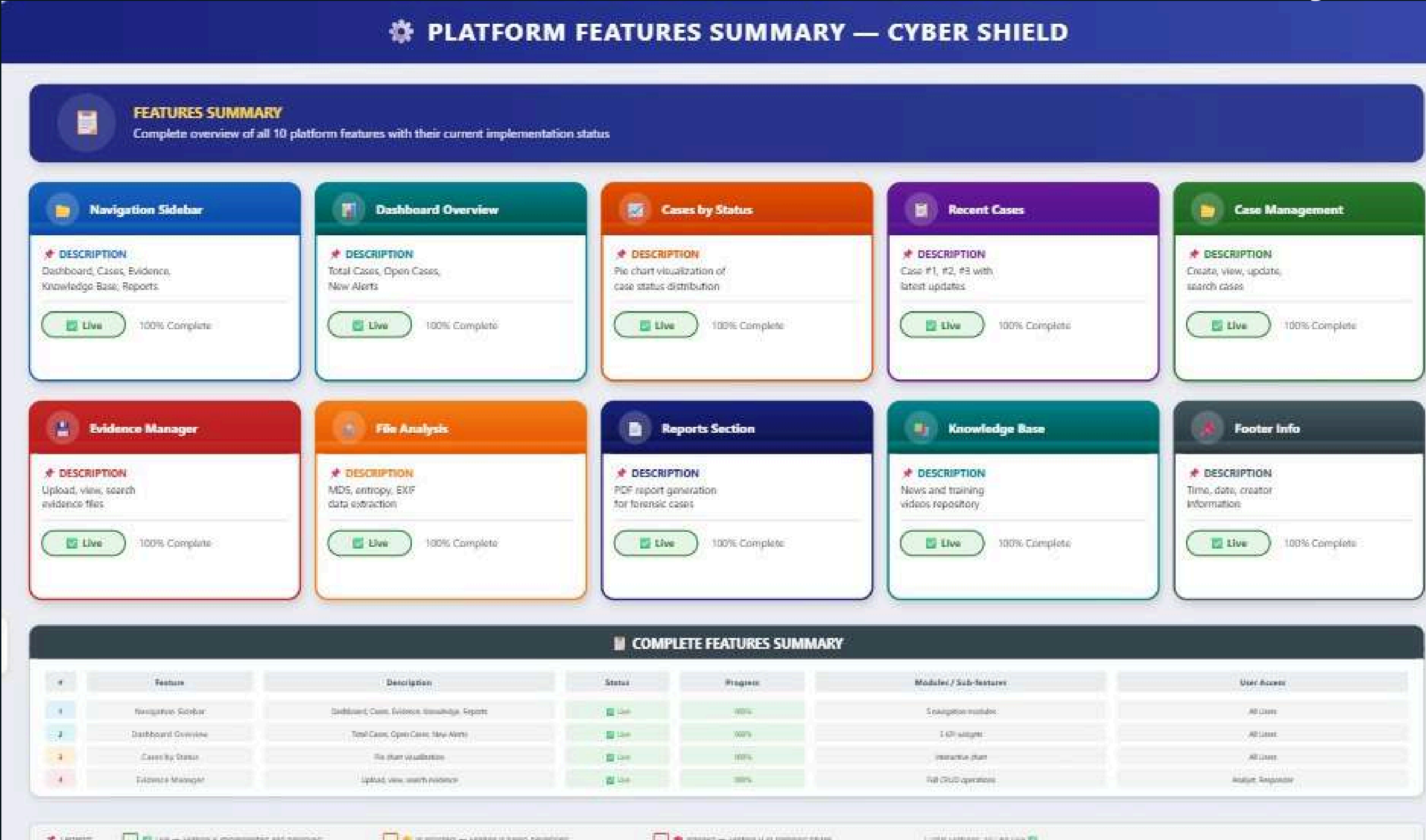
Cyber Shield represents a significant advancement in cybersecurity forensics platforms. By addressing the core challenges of case management, evidence handling, and knowledge integration, it provides a robust solution for modern security operations. The platform's user-centric design, comprehensive feature set, and scalability make it an invaluable tool for organizations of all sizes in their fight against cyber threats.

The successful implementation of Cyber Shield will contribute to :

- Enhanced organizational security posture
- Improved incident response capabilities
- Better regulatory compliance
- More efficient forensic investigations
- Higher stakeholder confidence



Appendix : Platform Features Summary :



Thank You !

In the battle against cyber threats, knowledge is our
weapon, and preparedness is our shield –
Cyber Shield stands guard at the intersection of innovation
and security.